FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS

FULL TEST– I
(PAPER-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	A
2.	D	B	B
3.	D	A	C
4.	C	C	A
5.	A, D	A, B, D	A, C
6.	B, D	A, B, C	A, B, C, D
7.	A, C	B, D	A, C, D
8.	B, C, D	B, C, D	A, B, C, D
9.	B	C	A
10.	D	D	C
11.	B	A	B
1.	(A) → (q) (B) → (p, q, s, t) (C) → (p, q, s, t) (D) → (r, s)	(A → s, t) (B → p, q, r, t) (C → p, s) (D → q, s, t)	(A) → (s) (B) → (q) (C) → (s) (D) → (p, r, t)
2.	(A) → (t) (B) → (q, s) (C) → (p, q, r) (D) → (t)	(A → q) (B → p, t) (C → p, q) (D → s)	(A) → (p) (B) → (q) (C) → (p) (D) → (r)
1.	1	4	3
2.	1	7	4
3.	2	3	3
4.	8	5	5
5.	2	2	4
6.	3	1	3

Physics

PART – I

SECTION – A

3. $\vec{F}_{net} = \frac{d\vec{p}}{dt} = 15\hat{i} - 20t\hat{j} - 50\hat{k} = \vec{F}_{real} + \vec{F}_{pseudo}$ and $\vec{F}_{real} = mg(-\hat{k}) = -50\hat{k}$ N

4. $Q = KA\frac{dT}{dx}$

$$\frac{Q}{A}\int_0^x dx = K_0\int_0^T (1+T)dT$$

$$\frac{Q}{A}x = K_0\left(T + \frac{T^2}{2}\right)_0^T$$

by solving

$$\frac{Q}{A}x = K_0\left(T + \frac{T^2}{2}\right)$$

So, $\frac{Q}{A}x_0 = K_0\left(300 + \frac{(300)^2}{2}\right)$

So, at $x = 2x_0$ temperature $T \approx 425$ K

5. $m = V\rho = \left[A_1 + A_2 + \sqrt{A_1A_2}\right]H\rho$

$$\rho H = \frac{m}{\left[A_1 + A_2 + \sqrt{A_1A_2}\right]}$$

$A_2\uparrow \Rightarrow \rho H \downarrow$

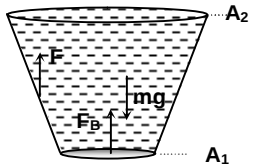
$P_1 = \rho_1 g H_1$

$P_2 = \rho_2 g H_2 \Rightarrow P_1 > P_2$

$F = mg - F_B = mg - PA_1$

$F_1 = mg - A_1P_1$,

$F_2 = mg - A_1P_2$



6. $U = \frac{-2GMm}{\sqrt{d^2 + y^2}} + \frac{-GM^2}{2d}$

$$= \frac{-2GMm}{\sqrt{d^2 + y^2}} - \frac{2Gm^2}{2d}$$

$$= -2Gm^2\left[\frac{2}{\sqrt{d^2 + y^2}} + \frac{1}{d}\right]$$

$$= |\Delta U| = \frac{1}{2}mv^2$$

7. $v = \lambda f = \frac{h}{p} \times \frac{E}{h} = \frac{E}{P} = \frac{P}{2m} = \left(\frac{v_{Particle}}{2}\right)$

8. $y = 5e^{-(\sqrt{a}\,x+\sqrt{b}\,t)^2}$

09-11. h = 160 m

$t_{1_0} = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 160}{10}} = 4\sqrt{2}\,\text{sec}$

Velocity of system after collision in vertical direction, v_{10} = 30 m/s;
Velocity of system after collision in horizontal direction, $v_{2_0} = ¼ \times 40 = 10$ m/s

$v_{1_0}'^2 = 900 + 2 \times 10 \times 80 \Rightarrow v_{1_0}' = 50$ m/s

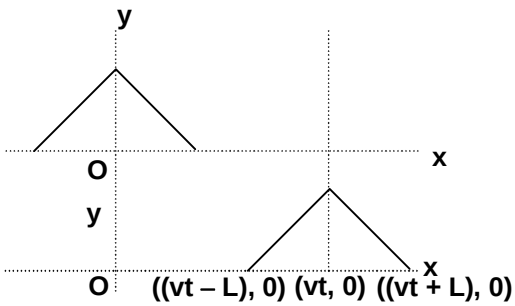
$t_{2_0} = \frac{50-30}{10} = 2$ sec

$t_{1_0} = 4$ sec

SECTION - B

2. Shape of sting at $t = 0$ is given by.

$$y(x,t) = \begin{cases} 0 & \text{if} \quad -\infty < x \le vt - L \\ h(L + x - vt)/L & \text{if} \quad vt - L \le x \le vt \\ h(L - x + vt)/L & \text{if} \quad vt \le x \le vt + L \\ 0 & \text{if} \quad vt + L \le x < \infty \end{cases}$$



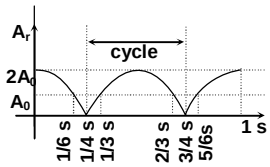
SECTION – C

2. $10g\sin\frac{\theta}{2} = 5g \sin\theta$

$10\sin\frac{\theta}{2} = 5\ (2 \sin\theta/2 \cos\theta/2)$

$\cos\theta/2 = 1$

4. $y_1 = A \sin\omega_1 t$
$y_2 = A \sin\omega_2 t$
$y_r = 2A\cos\left\{\frac{(\omega_2 - \omega_1)}{2}t\right\}\left\{\sin\frac{(\omega_2 + \omega_1)}{2}t\right\}$



Resultant amplitude $A_r = 2A|\cos(\Delta\omega)t/2|$

$(\Delta\omega)\frac{t}{2} = \frac{\pi}{2} \Rightarrow t = \frac{1}{4}$s

$(\Delta\omega)\frac{t}{2} = \frac{\pi}{3} \Rightarrow t = \frac{1}{6}$s

In one cycle of intensity of 1/2s, the detector remain idle for

$2\left(\frac{1}{4} - \frac{1}{6}\right)s = \frac{1}{6}\text{sec}$

$\therefore$ In ½ sec cycle, active time is $\left(\frac{1}{2}-\frac{1}{6}\right)$ = 1/3 sec

$\therefore$ In 12 sec interval, active time is $12\times\frac{(1/3)}{(1/2)}$ = 8 sec

6. $V_1=\frac{(2C)V}{3C}=\frac{2}{3}V$

$U_1=\frac{1}{2}C\frac{4}{9}V^2=\frac{2CV^2}{9}$

$V_2=\frac{V}{3}$

$U_2=\frac{1}{2}(2C)\frac{V^2}{9}=C\frac{V^2}{9}=\frac{CV^2}{9}$

Initial total energy = $\frac{CV^2}{3}$

Final total energy = $\frac{1}{2}(2C)V^2=CV^2$

Charge flow through the battery $=2CV-\frac{2}{3}CV=\frac{4}{3}CV$

$\Rightarrow\ W_{battery}=\frac{4}{3}CV^2$

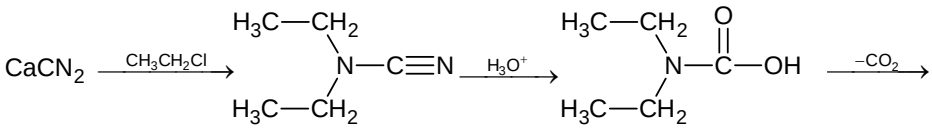
$\Rightarrow$ Heat dissipated $=\frac{4CV^2}{3}-\frac{2CV^2}{3}=\frac{2}{3}CV^2$

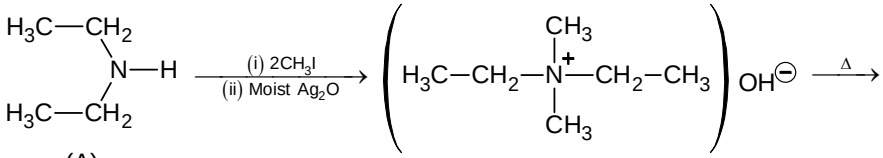
Chemistry

PART – II

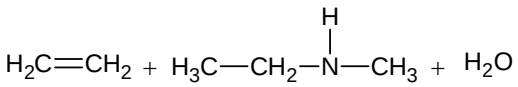
SECTION – A

1.



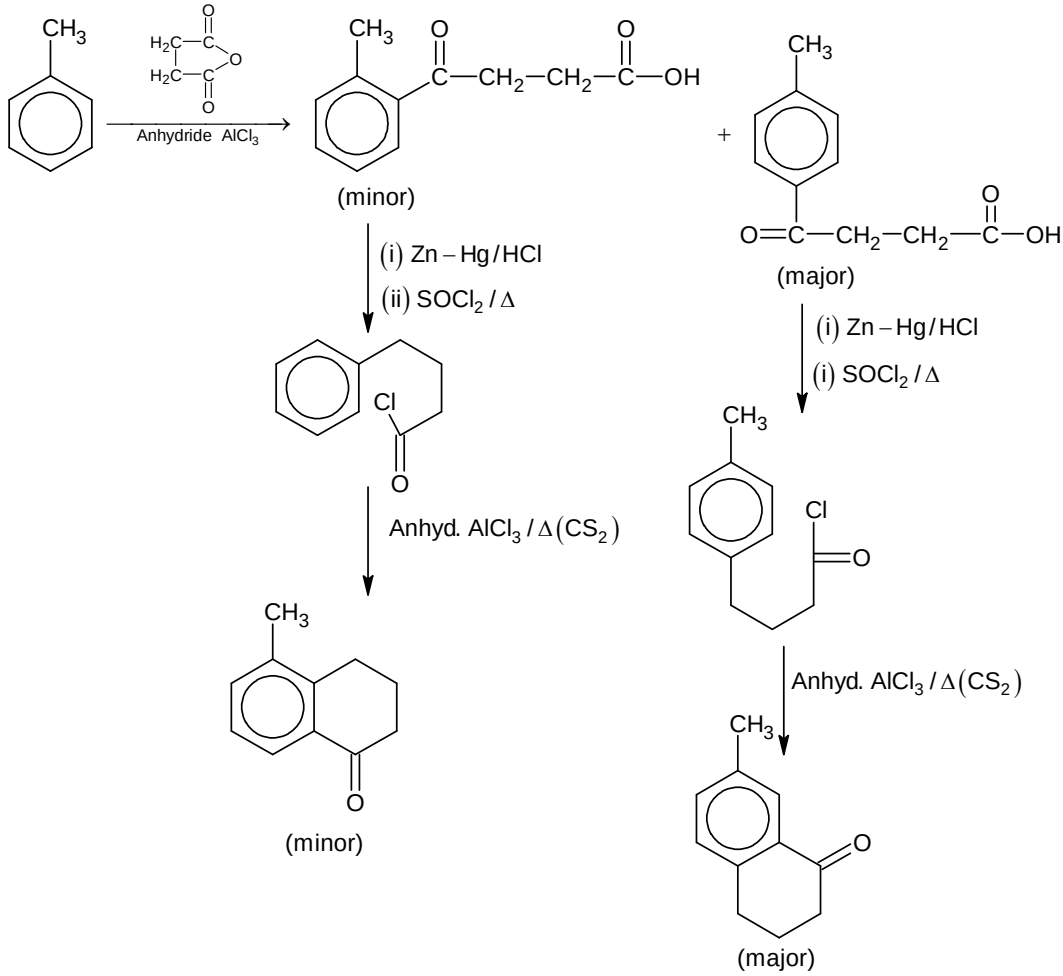


(A) (B)

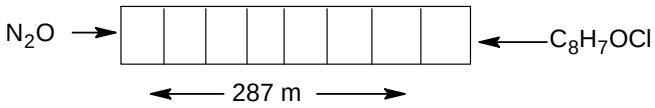


(C)

2.



3.



Molar weight of $C_8H_7OCl = 8 \times 12 + 7 \times 1 + 16 + 35.5$
$= 154.5$ g

Molar weight of $N_2O = 2 \times 14 + 16 = 44$ g

According to Graham's Law of diffusion

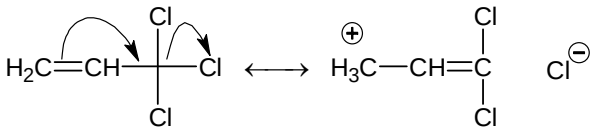
$$\frac{r_{N_2O}}{r_{C_8H_7OCl}} = \sqrt{\frac{M_{C_8H_7OCl}}{M_{N_2O}}} = \sqrt{\frac{154.5}{44}} = \sqrt{3.5} = 1.87 : 1$$

$$d_{N_2O} = \frac{1.87}{2.87} \times 287 = 187^{th} \text{ row from } N_2O \text{ side.}$$

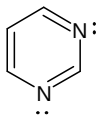
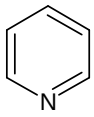
$$d_{C_8H_7OCl} = \frac{1.0}{2.87} \times 287 = 100^{th} \text{ row from weeping gas side.}$$

Therefore, the spectator from the side of N_2O in the 187th. Row will be laughing and weeping simultaneously. Alternatively, the spectator from the side of weeping gas in 100th row will laugh and weep.

4. The given orbital diagram has two radial nodes, i.e. $n - \ell - 1 = 2$
$\therefore$ The orbital, in which electron is present is 3s.

5. (A) 

These occurs decrease in electron density and hence the ionization potential increases.

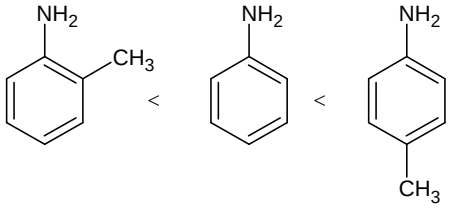
(B) Basic strength $\propto$ electron density over one N-atom  has lesser electron density over one N-atom than in 

(C) Heat of combustion $\propto \frac{1}{\text{Stability}}$

stability $\propto$ No. of α –H

and if no. of α–H of same stability $\propto$ ease of hyperconjugation.

(D) Conjugate acid of weaker base is stronger acid. Basic strength



6. B. pt. of CCl_4 < that of $SiCl_4$. The vapour must always be richer in more volatile component. Hence, the upper curve represents the composition of the vapour and the lower curve represents the composition of the solution at corresponding B. pt.

8. The correct basic order for (a) option

$$CH_3O^{\ominus} > CH_3NH^{\ominus} > CH_3CH_2^{\ominus}$$

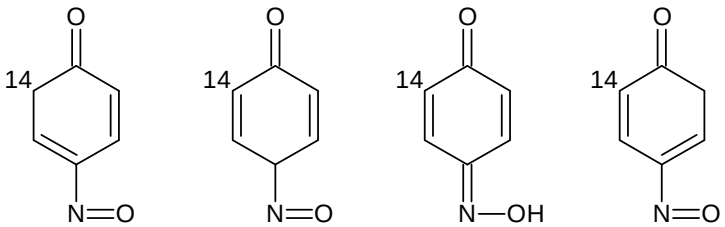
(-ve) charge on most electronegative atom — (-ve) charge on least electronegative atom

and it can also explain by its conjugate rest all option are correct.

Solution for the Q. No. 9 to 11.

M – $PbCO_3$
P – $Pb(NO_3)_2$
B – PbO
C – CO_2
D – NO_2
E – O_2
F – $CaCO_3$
G – $Ca(HCO_3)_2$
Q – $PbSO_4$
R – $NaNO_3$

SECTION –C

1. 

2. $Z_{B^-} = 4 - 2 = 2$

$Z_{A^+} = 4$

$Z_{C^{-2}} = 1$

$\therefore A_4B_2Z_1$

x + y + z = 7

3. $\pi r^2 = 5.49\ \overset{o}{A}^2$

$2\pi r = 83.05\ \overset{o}{A}$

then $r = 13.22\ \overset{o}{A}$

$\because\ r = \dfrac{0.529 \times n^2}{Z}$

$\therefore\ 13.22 = \dfrac{0.529 \times n^2}{Z}$

n = 5

Number of maximum lines possible = 3

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

4.

Int. No. of mole	10^{-6} V	10^{-6} V	0	0
at t = t sec	0	0	10^{-6} V	

$[CH_3COONa] = \dfrac{10^{-6}V}{2V}$

$= 5 \times 10^{-7} < 10^{-6}$.

$\left[OH^-\right] = \sqrt{\dfrac{K_w \times C}{K_a}}$

$= \sqrt{\dfrac{10^{-4} \times 10^{-7} \times 5}{1.9 \times 10^{-5}}}$

$= 2.63 \times 10^{-8} < 10^{-6}$

so pOH = –log $(2.63 \times 10^{-8} + 10^{-7})$

pH = 7

$\Rightarrow \dfrac{pH}{1.42} = 5$

5. $Mn^{+2} \longrightarrow MnO_2 + 2e^-$

$4e^- + O_2 \longrightarrow 2H_2O$

1 Eq of Mn^{+2} = 1 Eq of O_2

(n = 2) (n = 4)

½ mole of Mn^{+2} = 1 Eq. of O_2

2 mole of Mn^{+2} = 4 Eq. of O_2

6. $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$ $\quad Keq = 10^{-3}$

$CO_2(g) \rightleftharpoons C(s) + O_2(g)$ $\quad Keq = 10^{2}$

$C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ $\quad Keq = 10^{-1} = X$

$C(s) + O_2(g) \rightleftharpoons 2C(s) + CO_2(g)$ $\quad Keq = 1/x = 10^{-1}$

$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$ $\quad Keq = 10^{-1} = Y$

Hence X : Y = 1 : 1

Mathematics

PART – III

SECTION – A

1. Let z_1, z_2 be the two roots with $|z_1| = 1$

$\because z_1z_2 = \frac{c}{a} \Rightarrow |z_2| = \left|\frac{c}{a}\right|\frac{1}{|z_1|} = 1$

$\Rightarrow z_1\bar{z}_1 = z_2\bar{z}_2 = 1$

$\because z_1 + z_2 = -\frac{b}{a}$ and $|b| = |a|$

$\Rightarrow |z_1 + z_2|^2 = 1$

$\Rightarrow (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = 1 \Rightarrow (z_1 + z_2)\left(\frac{1}{z_1} + \frac{1}{z_2}\right) = 1$

$\Rightarrow (z_1 + z_2)^2 = z_1z_2 \Rightarrow b^2 = ac$

2. For $x \in [0, 1]$, $f'(x) \geq f'(1)$ ($\because$ $f'(x)$ is decreasing)

$\Rightarrow \frac{f'(x)}{f^2(x)+1} \geq \frac{f'(1)}{f^2(x)+1}$

$\Rightarrow \int_0^1 \frac{f'(x)}{f^2(x)+1}dx \geq f'(1)\int_0^1 \frac{dx}{f^2(x)+1}$

$\Rightarrow \tan^{-1}f(x)\Big|_0^1 \geq f'(1)\int_0^1 \frac{dx}{f^2(x)+1}$

$\Rightarrow \int_0^1 \frac{dx}{f^2(x)+1} \leq \frac{\tan^{-1}f(1)}{f'(1)}$

$\because \tan^{-1}\alpha < \alpha \ \forall\ \alpha > 0 \Rightarrow \frac{\tan^{-1}f(1)}{f'(1)} < \frac{f(1)}{f'(1)}$

$\Rightarrow \int_0^1 \frac{dx}{f^2(x)+1} < \frac{f(1)}{f'(1)}$

3. $\because$ P(x) = 0 has n real roots

$\Rightarrow$ x P(x) = 0 has n + 1 real roots

$\Rightarrow \frac{d}{dx}\left(xP(x)\right) = 0$

$\Rightarrow$ x P′(x) + P(x) = 0 has n real roots

$\Rightarrow$ x(x P′(x) + P(x)) = 0 has n + 1 real roots

$\Rightarrow \frac{d}{dx}\left(x^2P'(x) + xP(x)\right) = 0$ has n real roots

i.e. x^2 P″(x) + 3x P′(x) + P(x) = 0 has n real roots $\Rightarrow$ (I) is true

Let one of the roots of P(x) = 0 is α, and one root of Q′(x) = 0 is β as given, $\alpha \neq \beta$

$\because$ P′(x) Q′(x) + P(x) Q″(x) = 0

$\Rightarrow \frac{d}{dx}\left(P(x)Q'(x)\right) = 0$

Consider f(x) = P(x) Q′(x)

f(α) = 0, f(β) = 0

Hence, by Rolles theorem $\frac{d}{dx}\left(P(x)Q'(x)\right) = 0$ has at least one real root lying between α and β

4. **Case-I:**

If a $\in$ [0, 1], the curves intersect at $\left(\frac{a}{3}, \frac{a}{3}\right)$ and (a, a). The bounded region is contained in the triangle with vertices (0, 0), $\left(\frac{1}{2}, 0\right)$ and (1, 1) with area = $\frac{1}{4}$

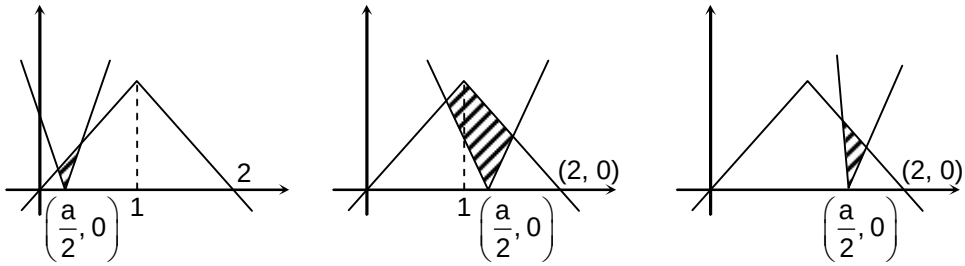
Hence, area can not exceed $\frac{1}{4}$

Case-II:

If a $\in$ [1, 3]. In this case the bounded region is a quadrilateral with four vertices $\left(\frac{a}{3}, \frac{a}{3}\right)$, $\left(\frac{a}{2}, 0\right)$, $\left(\frac{a+2}{3}, \frac{4-a}{3}\right)$ and (1, 1). In this case area bounded = $1 - \frac{1}{2}\cdot\frac{a}{3}\cdot\frac{a}{2} - \frac{1}{2}\left(\frac{4-a}{3}\right)\left(2-\frac{a}{2}\right)$

= $\frac{1}{3} - \frac{(a-2)^2}{6} \leq \frac{1}{3}$

Case-III: If a $\in$ [3, 4]. This case is symmetric with case-I



5. Multiplying the first DE by gh, the second by fh and the third by fg, and adding the equations gives

(fgh)′ = 6(fgh)^2 + 6

Let f(x) g(x) h(x) = k(x)

We have k′(x) = 6(k(x))^2 + 6

Integrating and using k(0) = 1 gives

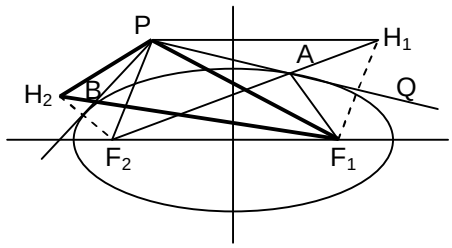
k(x) = $\tan\left(6x + \frac{\pi}{4}\right)$

Now 1st DE $\Rightarrow \dfrac{f'(x)}{f(x)} = 2k(x) + \dfrac{1}{k(x)}$

$\Rightarrow \dfrac{f'(x)}{f(x)} = 2\tan\left(6x + \dfrac{\pi}{4}\right) + \cot\left(6x + \dfrac{\pi}{4}\right)$

Integrating and using f(0) = 1 gives option A and C

6. By reflection property of ellipse $\angle PAF_2 = \angle QAF_1$
$\because$ H_1 is reflection of F_1 $\therefore$ $\angle QAH_1 = \angle QAF_1$
Now $PH_2 = PF_2$, $PH_1 = PF_1$
and $F_2H_1 = F_2A + AF_1 = 2a$
and $F_1H_2 = F_1B + BH_2 = F_1B + F_2B = 2a$
Hence, ΔPF_2H_1 and ΔPF_1H_2 are congruent as all three sides are equal
This leads to all the options



7. Let $f(x) = a(x - \alpha_1)(x - \beta_1)$
$\Rightarrow f(x)\, f(x+1) = a^2(x - \alpha_1)(x - \beta_1 + 1)(x - \beta_1)(x - \alpha_1 + 1)$
$= a^2(x^2 + x - \alpha_1 x - \beta_1 x + \alpha_1\beta_1 - \alpha_1)(x^2 + x - \alpha_1 x - \beta_1 x + \alpha_1\beta_1 - \beta_1)$
$\Rightarrow f(x)\, f(x+1) = a^2[(x^2 - (\alpha_1 + \beta_1 - 1)\, x + \alpha_1\beta_1) - \alpha_1] \times [(x^2 - (\alpha_1 + \beta_1 - 1)\, x + \alpha_1\beta_1) - \beta_1]$
Hence, $g(x) = a^2(x - \alpha_1)(x - \beta_1)$ and $h(x) = x^2 - (\alpha_1 + \beta_1 - 1) + \alpha_1\beta_1$

8. Consider $h(x) = x(1 + \ln x)$, $h: [1, \infty) \to [1, \infty)$
Clearly h(x) to increasing function in its domain
$\because$ $f(\lambda)$ denotes the solution of $x(1 + \ln x) = \lambda$
$\therefore$ $f: [1, \infty) \to [1, \infty)$, f is inverse function of h(x)
As $\lim\limits_{\lambda\to\infty} h(\lambda) = \infty \Rightarrow \lim\limits_{\lambda\to\infty} f(\lambda) = \infty$

$$f'(\lambda) = \frac{1}{h'(x)\big|_{x=f(\lambda)}} \Rightarrow f'(\lambda) = \frac{1}{2 + \ln x}\bigg|_{x=f(\lambda)} = \frac{1}{2 + \ln(f(\lambda))}$$

Now consider $L = \lim\limits_{\lambda\to\infty} \dfrac{f(\lambda)\ln\lambda}{\lambda} = \lim\limits_{\lambda\to\infty} \dfrac{f(\lambda)\ln\lambda}{f(\lambda)(1 + \ln(f(\lambda)))} = \lim\limits_{\lambda\to\infty} \dfrac{\ln\lambda}{1 + \ln(f(\lambda))}$

By L'Hospitals Rule $L = \lim\limits_{\lambda\to\infty} \dfrac{1/\lambda}{f'(\lambda)f(\lambda)} = \lim\limits_{\lambda\to\infty} \dfrac{f(\lambda)}{\lambda f'(\lambda)} = \lim\limits_{\lambda\to\infty} \dfrac{2 + \ln f(\lambda)}{1 + \ln f(\lambda)}$

$$\Rightarrow L = \lim_{\lambda\to\infty}\left(\frac{\dfrac{2}{\ln f(\lambda)} + 1}{\dfrac{1}{\ln f(\lambda)} + 1}\right) = 1$$

9. (I) holds true obviously
For (II): $\vec{A}(\vec{B} + \vec{C}) = (a_1, a_2)(b_1 + c_1, b_2 + c_2)$
$= 2a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_1(b_2 + c_2) + a_2(b_1 + c_1)$
$= (2a_1b_1 + a_2b_2 + a_1b_2 + a_2b_1) + (2a_1c_1 + a_2c_2 + a_1c_2 + a_2c_1)$
$= \vec{A}\cdot\vec{B} + \vec{A}\cdot\vec{C}$
Similarly it can be checked that (III) also holds true

10. $\vec{A}\cdot\vec{B} = \sqrt{a_1b_1} + \sqrt{a_2b_2} + 2\sqrt{a_2b_1} + 2\sqrt{a_1b_2}$

$\Rightarrow \vec{A}\cdot\vec{B} \le \left(\frac{a_1+b_1}{2}\right)+\left(\frac{a_2+b_2}{2}\right)+(a_2+b_1)+(a_1+b_2) \le \frac{3}{2}\left((a_1+a_2)+(b_1+b_2)\right)$

$\Rightarrow \vec{A}\cdot\vec{B} \le \frac{3}{2}\left(|\vec{A}|+|\vec{B}|\right)$

11. $\vec{A}\cdot\vec{B} = |a_1b_1 + a_2b_2 + a_3b_3| = \vec{B}\cdot\vec{A}$

Hence, (I) holds true

$\vec{A}(\vec{B}+\vec{C}) = |a_1(b_1+c_1)+a_2(b_2+c_2)+a_3(b_3+c_3)| = |(a_1b_1+a_2b_2+a_3b_3)+(a_1c_1+a_2c_2+a_3c_3)|$

$\neq |a_1b_1+a_2b_2+a_3b_3|+|a_1c_1+a_2c_2+a_3c_3|$

$\Rightarrow$ (II) Don't hold true

Similarly (III) can be checked

SECTION – B

1. Normal $y + tx = at^3 + 2at$

$\Rightarrow at^3 + (2a - h)t - k = 0$

$\Rightarrow t_1 + t_2 + t_3 = 0 \Rightarrow t_3 = -(t_1 + t_2)$ ..... (1)

$\Rightarrow t_1t_2 + t_2t_3 + t_3t_1 = \frac{2a-h}{a}$ ..... (2)

Also, $t_1t_2t_3 = \frac{k}{a}$ ..... (3)

$\Rightarrow \Delta ABC$ is equilateral

So, $G \equiv H \equiv O$

$G \equiv \left(\frac{a}{3}(t_1t_2+t_2t_3+t_3t_1), \frac{a}{3}(t_1+t_2+t_2+t_3+t_3+t_1)\right)$

$G \equiv \left(\frac{a}{3}(t_1t_2+t_2t_3+t_3t_1), 0\right)$ ..... (4)

$H \equiv \left(-a, \frac{a}{3}(t_1+t_2+t_3+t_1t_2t_3)\right)$ ..... (5)

Comparing (4) and (5), $t_1t_2t_3 = 0$ and from (3), $k = 0$

From equation (2), we get $\frac{2a-h}{a}\times\frac{a}{3} = -a$

$2a - h = -3a$

$h = 5a$

$\Rightarrow O \equiv$ circumcentre of $\Delta ABC = (-a, 0)$

Image of O will lie on circle

$\therefore N \equiv (5a, 0)$

$\Rightarrow$ One of the intersection point lie on vertex ($\because$ k = 0)

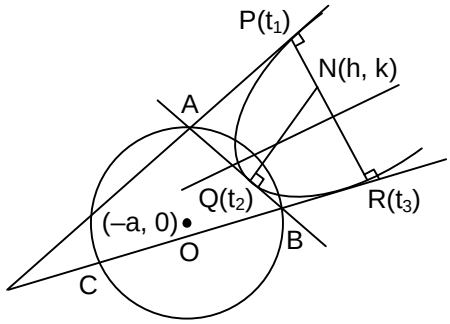
So, $t_2 = 0$

$\Rightarrow at^3 + (2a - 5a)t = 0$

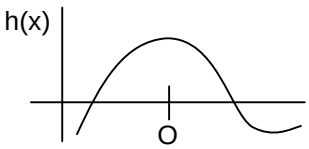
$\Rightarrow t^3 - 3t = 0 \Rightarrow t(t^2 - 3) = 0$

$t = 0, \pm\sqrt{3}$

$\therefore$ Intersection points $\Rightarrow P \equiv \left(3a, 2a\sqrt{3}\right)$, $Q \equiv (0, 0)$, $R \equiv \left(3a, -2a\sqrt{3}\right)$



2. (A) $(f(x))^2 + [f'(x)]^2 = h(x)$



$h'(x) = 2f(x)\, f'(x) + 2f'(x)\, f''(x)$

$= 2f'(x)\,(f(x) + f''(x))$

$= 2f'(x)\,\{-x\, g(x)\, f'(x)\}$

$= -2x(f'(x))^2\, g(x)$

So, $h'(x) < 0$ of $x > 0$ and $h'(x) > 0$ for $x < 0$

Then, $x = 0$ is a maxima

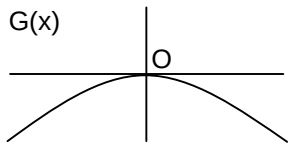
(B) $f^2(x) + f(x) = g(x)$

$2f(x)\, f'(x) + f'(x) = g'(x)$ and $g'(x) = f'(x)\,\{2f(x) + 1\}$

So, $g'(x) \geq 0$ as $g(x)$ is increasing and for $f'(x) \geq 0$

$2f(x) + 1 \geq 0$

Hence, $f(x) \geq -\frac{1}{2}$



(C) Let $G(x) = \left[\int_0^x f(t)\,dt\right]^2$ ..... (1)

$G'(x) = 2f(x)\int_0^x f(t)\,dt$ and $\frac{G'(x)}{2} = g(x)$

and $G'(0) = 0 = g(0)$

Also, $g(x)$ is non-increasing

Hence, $g(x) \geq 0\ x \in (-\infty, 0)$, $g(x) \leq 0\ x \in [0, \infty)$

i.e., $G'(x) \geq 0$ for $x \leq 0$, $G'(x) \leq 0$ for $x \geq 0$ and $G(0) = 0$

But $G(x)$ (from equation (1)) ≥ 0

Hence, $G(x) = 0\ x \in R$

$\int_0^x f(t)\,dt = 0\ \ x \in R$

i.e., $f(x) = 0\ x \in R$

Hence, $f(1) = 0$

(D) $f(x) = x^2 + \int_0^x e^{-t} f(x-t)\,dt$ ..... (1)

$f(x) = x^2 + e^{-x}\int_0^x e^{t} f(x-x+t)\,dt$

$f(x) = x^2 + e^{-x}\int_0^x e^{t} f(t)\,dt$

$f'(x) = 2x + e^{-x}\left[e^{x} f(x)\right] - e^{-x}\int_0^x e^{t} f(t)\,dt$

$f'(x) = 2x + f(x) - \int_0^x e^{-(x-t)} f(t)\,dt$ $= 2x + f(x) + x^2 - f(x)$

$f'(x) = 2x + x^2$

$f(x) = x^2 + \frac{x^3}{3} + c$ and from equation (1) $f(0) = 0$

Hence, $c = 0$

i.e., $f(x) = x^2 + \frac{x^3}{3}$

$f(-1) = 1 - \frac{1}{3} = \frac{2}{3}$, $\frac{k}{3} = \frac{2}{3}$ $\therefore\ k = 2$

SECTION – C

1. Let $f(r) = \int_0^{\pi/2} x^r \sin x dx$

Now, $\int_0^{\pi/2} x^r \cos x dx = \frac{x^{r+1}}{r+1}\cos x\Big|_0^{\pi/2} + \int_0^{\pi/2} \frac{x^{r+1}}{r+1}\sin x dx$

$\Rightarrow \int_0^{\pi/2} x^r \cos x dx = \frac{f(r+1)}{(r+1)}$

So, we have $\lim_{r\to\infty} \frac{r^C f(r)(r+1)}{f(r+1)} = L$

Now, consider $f(r) < \int_0^{\pi/2} x^r dx = \frac{\left(\frac{\pi}{2}\right)^{r+1}}{r+1}$

Also as $\sin x > \frac{2x}{\pi} \ \forall\ x \in \left(0, \frac{\pi}{2}\right)$

$f(r) > \int_0^{\pi/2} \frac{2}{\pi} x^{r+1} dx = \frac{\left(\frac{\pi}{2}\right)^{r+1}}{r+2}$

$\Rightarrow \frac{r}{r+2} < r\left(\frac{2}{\pi}\right)^{r+1} f(r) < \frac{r}{r+1}$

Hence, $\lim_{r\to\infty} \frac{f(r)}{f(r+1)} = \lim_{r\to\infty} \frac{r\left(\frac{2}{\pi}\right)^{r+1} f(r)}{(r+1)\left(\frac{2}{\pi}\right)^{r+2} f(r+1)} \times \frac{2(r+1)}{\pi r} = \frac{2}{\pi}$

Now, $\lim_{r\to\infty} r^C (r+1)\frac{f(r)}{f(r+1)} = L \Rightarrow \lim_{r\to\infty} \frac{2}{\pi} r^C (r+1) = L$

For positive L we should have C = –1 and L = $\frac{2}{\pi}$

2. Let x + y = α and x – y = $\beta \Rightarrow x = \frac{\alpha+\beta}{2}$ and $y = \frac{\alpha-\beta}{2}$

Given equation $\Rightarrow \beta f(\alpha) - \alpha f(\beta) = (\alpha^2 - \beta^2)\alpha\beta \Rightarrow \frac{f(\alpha)}{\alpha} - \alpha^2 = \frac{f(\beta)}{\beta} - \beta^2$

Hence, $\frac{f(x)}{x} - x^2 = k$ (k is some real constant)

$\Rightarrow f(x) = x^3 + kx$

$f(1) = -2 \Rightarrow k = -3, f(x) = x^3 - 3x$

$f'(x) = 3(x - 1)(x + 1)$ on the interval $x \in \left[-\sqrt{3}, \sqrt{3}\right]$ extreme value are –2 and 2

3. The given equation can be re-written as quadratic in y

$y^2 + (8x - 2x^2)y + (3x^2 + 6x) = 0$

Its discriminant to equal to

$= (64x^2 - 32x^3 + 4x^4) - 4(3x^2 + 6x) = 4x(x^3 - 8x^2 + 13x - 6)$
$= 4x(x - 6)(x - 1)^2$
For solutions in integers this discriminant should by perfect square this happens if and only if
$z^2 = x(x - 6) = (x - 3)^2 - 9$
$9 = (x - 3)^2 - z^2 = (x + z - 3)(x - z - 3)$ for name integer z
Checking all factorisations $9 = (-9) \times (-1) = (-3) \times (-3) = (-1) \times (-9) = 9 \times 1 = 3 \times 3 = 1 \times 9$
This gives all possible pairs (x, y) as (x, y) = (–2, 0), (–2, 24), (0, 0), (8, 4), (8, 60), (6, 12)
Therefore exactly three pairs of natural number (x, y) are possible

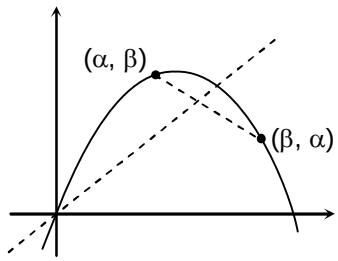
4. As no multiple of π nor any odd multiple of $\frac{\pi}{4}$ satisfies the equation we can multiply both sides by sin x cos 2x, which gives
$\sin 8x \cos 5x = 2 \sin 4x \cos 4x \cos 5x = 4 \sin 2x \cos 2x \cos 4x \sin 5x$
$= (\sin x \sin 2x)(8 \cos x \cos 4x \cos 5x) = \sin x \cos 2x$
$\Rightarrow \sin 13x + \sin 3x = \sin 3x - \sin x$
$\Rightarrow \sin 13x = \sin(-x)$
$\Rightarrow 13x = n\pi + (-1)^n(-x)$
Which gives 10 solutions $\frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7}, \frac{4\pi}{7}, \frac{5\pi}{7}, \frac{6\pi}{7}$

5. Let $H_i \equiv$ the event of getting a head on i^{th} coin
H ≡ getting odd number of heads on throwing n coins once

Coin Number:	1	2	3	4	……..	m	…..	n	
$P(H_i)$	$\frac{1}{3}$	$\frac{1}{5}$	$\frac{1}{7}$	$\frac{1}{9}$		$\frac{1}{(2m+1)}$	…..	$\frac{1}{(2n+1)}$	….. (1)
$P(\bar{H}_i)$	$\frac{2}{3}$	$\frac{4}{5}$	$\frac{6}{7}$	$\frac{8}{9}$	……..	$\frac{2m}{(2m+1)}$	…..	$\frac{2n}{(2n+1)}$	….. (2)

Now, P(H) = sum of series which has each term consisting of product of an odd number of terms from (1) and even number of terms from (2) such that total number of factors in each term is n
Consider the product $\left(\frac{2}{3}-\frac{1}{3}\right)\left(\frac{4}{5}-\frac{1}{5}\right)\left(\frac{6}{7}-\frac{1}{7}\right).....\left(\frac{2n}{2n+1}-\frac{1}{2n+1}\right)=\frac{1}{2n+1}$
$\text{LHS} = P(\bar{H}) - P(H) = \frac{1}{2n+1}$
Also, $P(\bar{H}) + P(H) = 1$
$\Rightarrow P(H) = \frac{n}{2n+1}$
$\Rightarrow \alpha = 1, \beta = 2, \gamma = 1, \alpha + \beta + \gamma = 4$

6. (α, β) and (β, α) will lie on some line $y = -x + \lambda$
Solving line and parabola $-(-x + \lambda) = ax - bx^2$
$\Rightarrow bx^2 - (1 + a)x + \lambda = 0$ ….. (1)
Again, $y = a(\lambda - y) - b(\lambda - y)^2$
$\Rightarrow by^2 + y(1 + a - 2b\lambda) + (b\lambda^2 - a\lambda) = 0$ ….. (2)
Both (1) and (2) has same roots which are α and β
Hence, $1 + a - 2b\lambda = -(1 + a) \Rightarrow \lambda = \frac{1+a}{b}$



Now, for α, β to exist disc of (1) > 0
$\Rightarrow (1 + a)^2 - 4b\lambda > 0$
$\Rightarrow (a - 3)(a + 1) > 0$
$\Rightarrow a \in (-\infty, -1) \cup (3, \infty)$